

Update of the track finding network

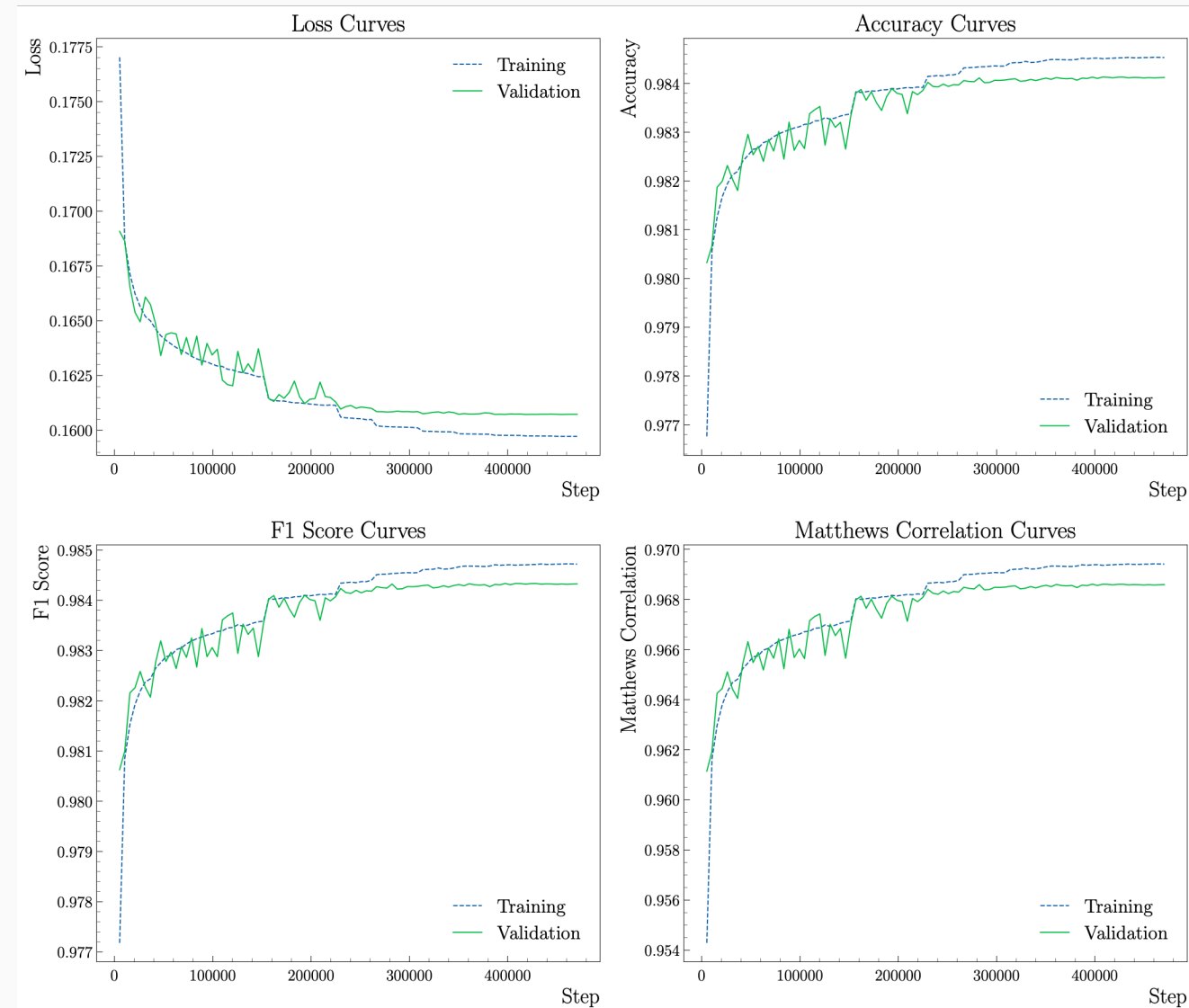
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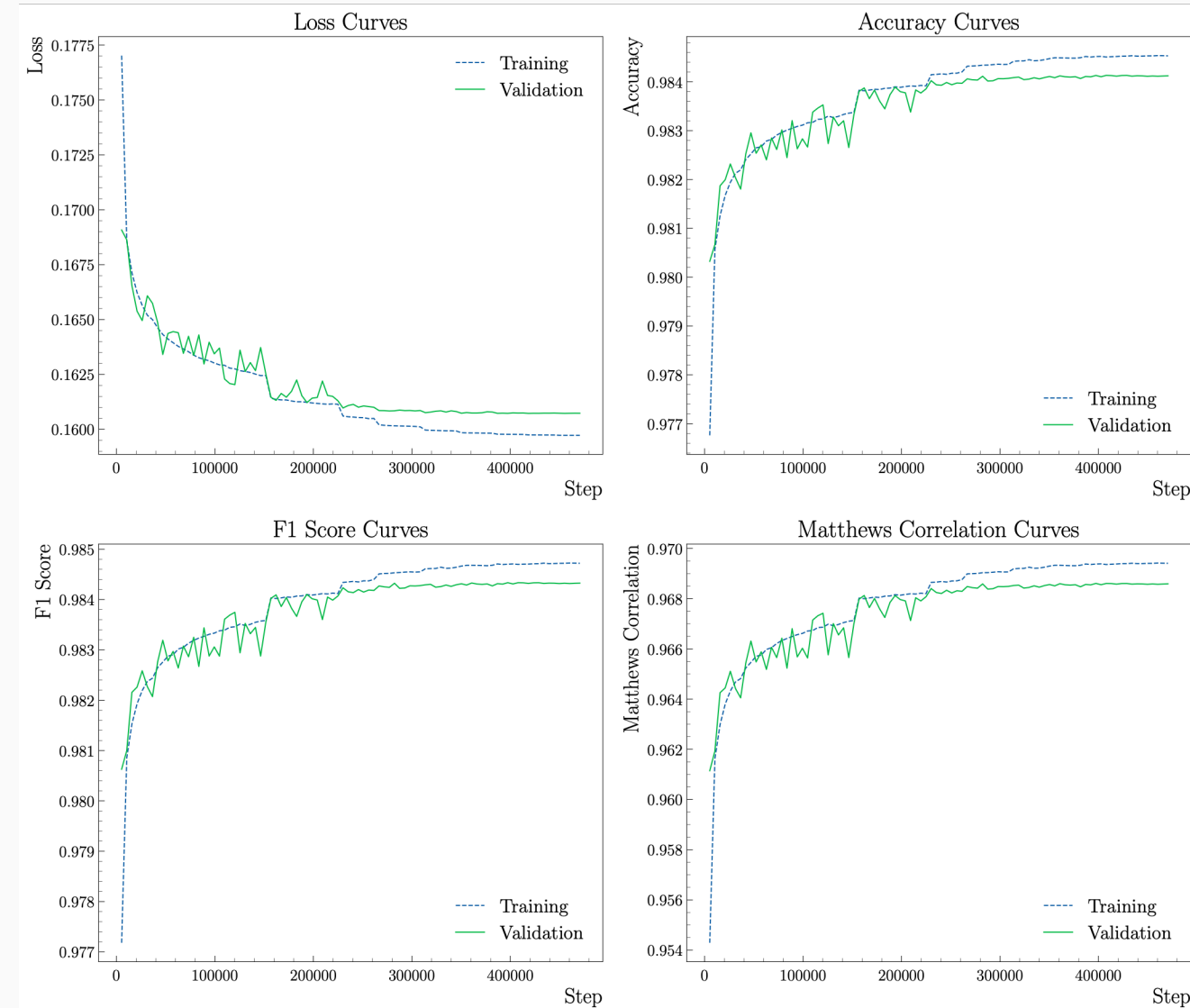
Model: Description and Training

- Model: MultiLayer Perceptron, 10 inputs, 1 output (size and number of layer have been optimized)
- Inputs: x and y values of the five inter-clusters
- For the training $\Rightarrow$ Need good and bad tracks:
 - Good tracks: GEANT4 simulation (particle with $p \in [0.07, 1.5]$ GeV/c, $\varphi \in [0, 360]^\circ$, $\theta \in [30, 150]^\circ$, and $V_z \in [-20, 20]$ cm)
 - False tracks: Interchanging randomly up to two inter-clusters with another event
 - Generate 5M events composed of all light nuclei (flat distribution for all variables)
 - Output: Number between 0 and 1, with 0/1 means bad/good track



Model: Description and Training

- Accuracy
 - Fraction of all predictions that are correct
 - $$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$
- F1 Score
 - Harmonic mean of precision and recall, providing a single metric that balances the ability to correctly identify true tracks (precision) with the ability to find all true tracks (recall).
 - $$F1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2, TP}{2, TP + FP + FN}$$
 - Ranges from 0 to 1, where 1 is perfect.
- Matthews Correlation Coefficient (MCC)
 - $$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$$
 - Ranges from -1 (total disagreement) to +1 (perfect prediction), with 0 meaning no better than random

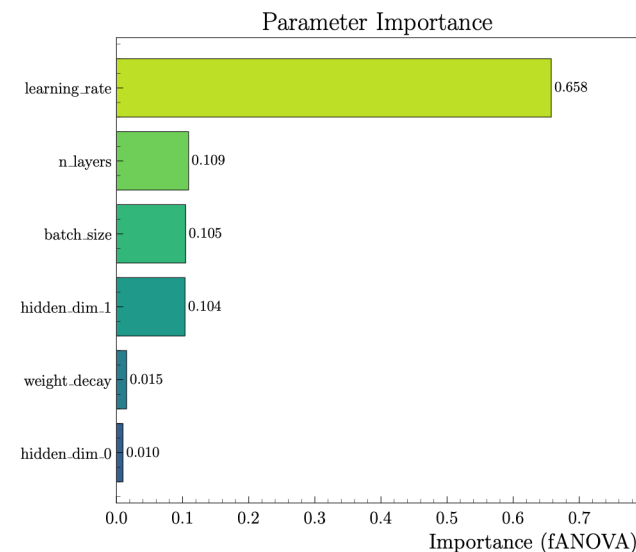
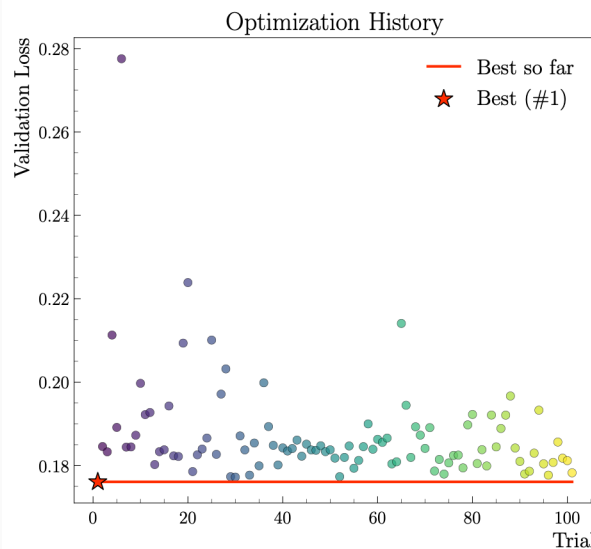




Model: Optimization

- Optuna framework with MedianPruner — prunes unpromising trials early
- Tunes: learning rate, weight decay, batch size, network depth, hidden dimensions

Hyperparameter Optimization Summary



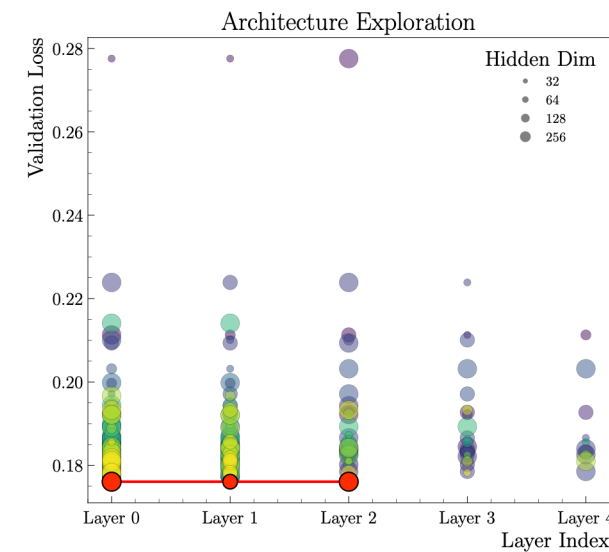
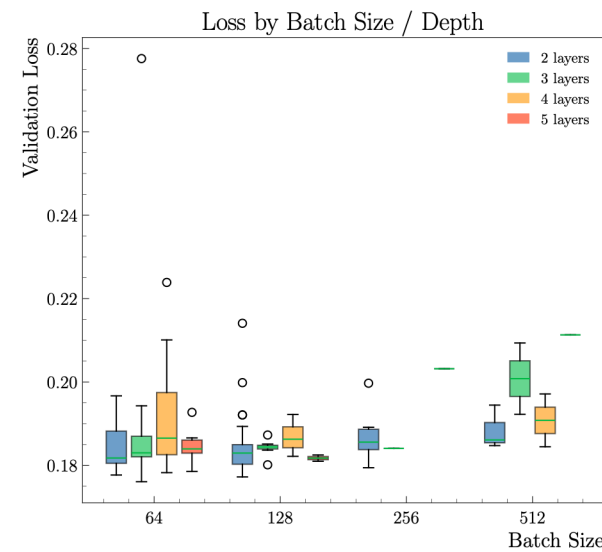
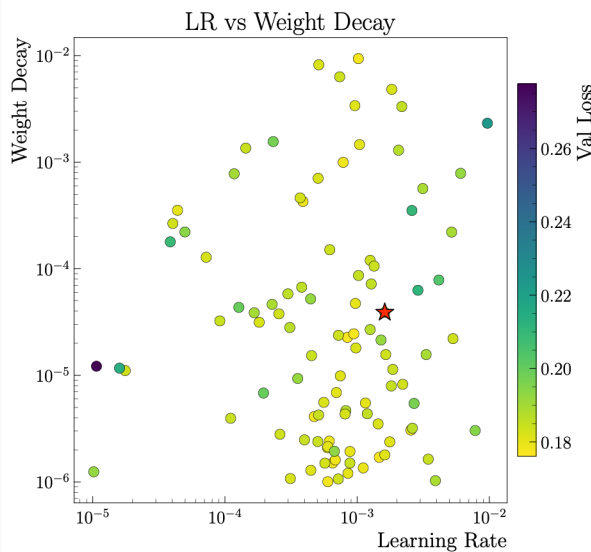
Best Configuration

Best Trial: #1
Val Loss: 0.17609

learning_rate: 1.62e-03
weight_decay: 3.90e-05
batch_size: 64
n_layers: 3

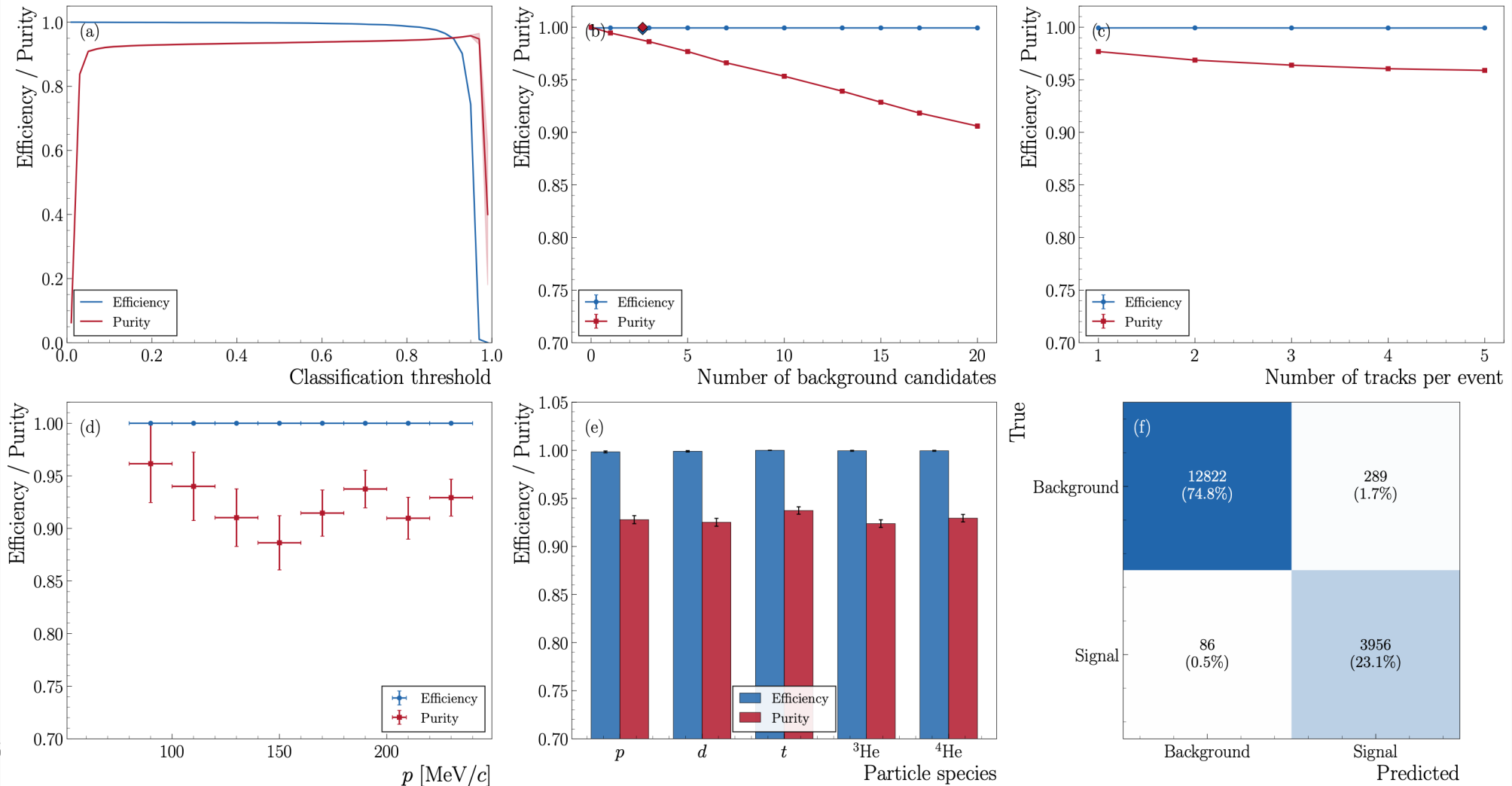
Architecture: 256 → 128 → 256

Trials: 101 done, 0 pruned, 1 failed



Evaluation on simulation

- Threshold sweep
- Noise: injects 0–20 random hits to measure degradation
- Per-momentum bin evaluation (60–240 MeV)
- Multi-track and particle-ID evaluation





Small test on data

- Just to see that we are not losing too much tracks
- Use 2.2 GeV proton run and check how many track are found by the two models

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Event-by-event comparison (50000 events)
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Events where new finds MORE tracks: 1132

Events where new finds FEWER tracks: 836

Events where new finds SAME tracks: 48032

Old total tracks: 10907 | New total tracks: 11365



Other

- The new model is in the networks repository:
 - <https://code.jlab.org/hallb/clas12/coatjava/networks>
- Not available in COATJAVA yet need a pull request
- Code to train and evaluated the network is available on GitLab:
 - <https://gitlab.com/MathieuOuillon/alert-track-finding>
- Next step: same things for track matching